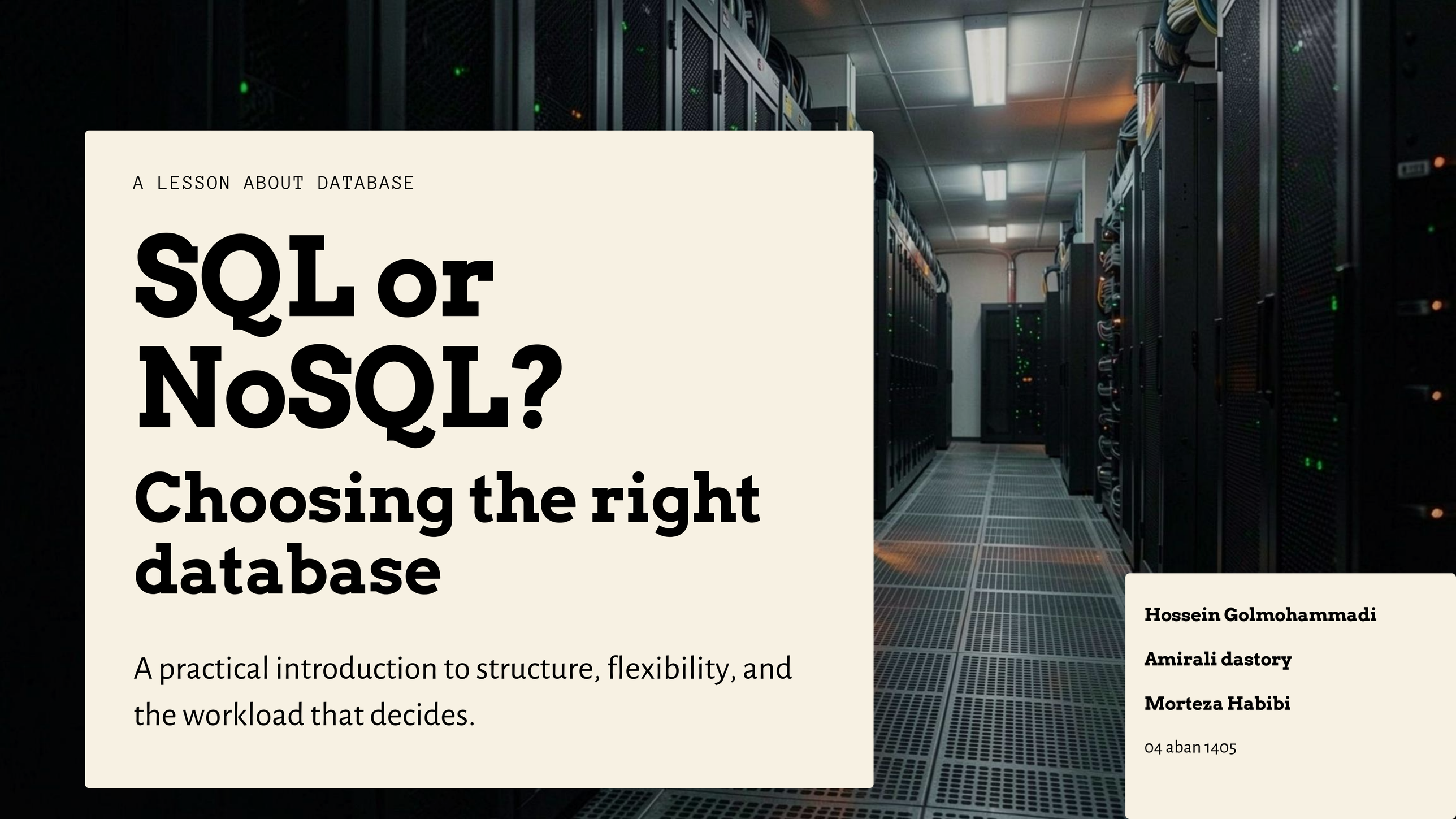


A LESSON ABOUT DATABASE

SQL or NoSQL?

Choosing the right database

A practical introduction to structure, flexibility, and the workload that decides.

Hossein Golmohammadi

Amirali dastory

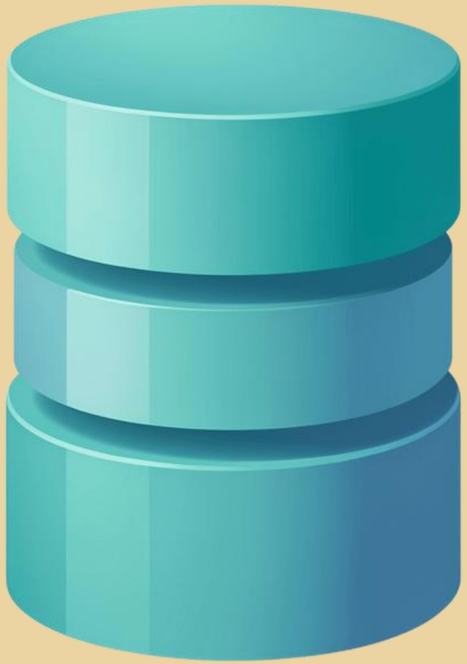
Morteza Habibi

04 aban 1405

START WITH THE JOB

A database is an apps organized memory

An application needs somewhere to store information, keep it organized, and retrieve it quickly.
When a user signs in, places an order, or opens a post, the app asks the database for the right pieces.



Primary keys and foreign keys

Students

StudentID (PK)	Name
1	Hossein
2	Amir
3	Ali

Courses

CourseID (PK)	Course	StudentID (FK)
C101	Intro to SQL	1
C102	Data Modeling	1
C103	Web Design	2

- **Primary key (PK):** the unique ID for each row. StudentID identifies one student; CourseID identifies one course. No two rows share the same value.
- **Foreign key (FK):** a column that points to another tables primary key. Courses.StudentID links each course to a student in the Students table, so you can see who is enrolled in what.

Primary key vs. foreign key

Feature	Primary key (PK)	Foreign key (FK)
Purpose	Identifies each row in its own table	Links a row to a row in another table
Uniqueness	Must be unique	Can repeat
Null values	Not allowed	Allowed when there is no link yet
Where it points	Stays in its own table	Points to a primary key in another table
Example	StudentID in Students	StudentID in Courses

Be careful

- A primary key must be unique and identify one row.
- A foreign key should match an existing primary key in the related table.
- Matching data types matter.

WHY SQL?

Why choose relational databases?

Accuracy

Useful when records must agree.

Relationships

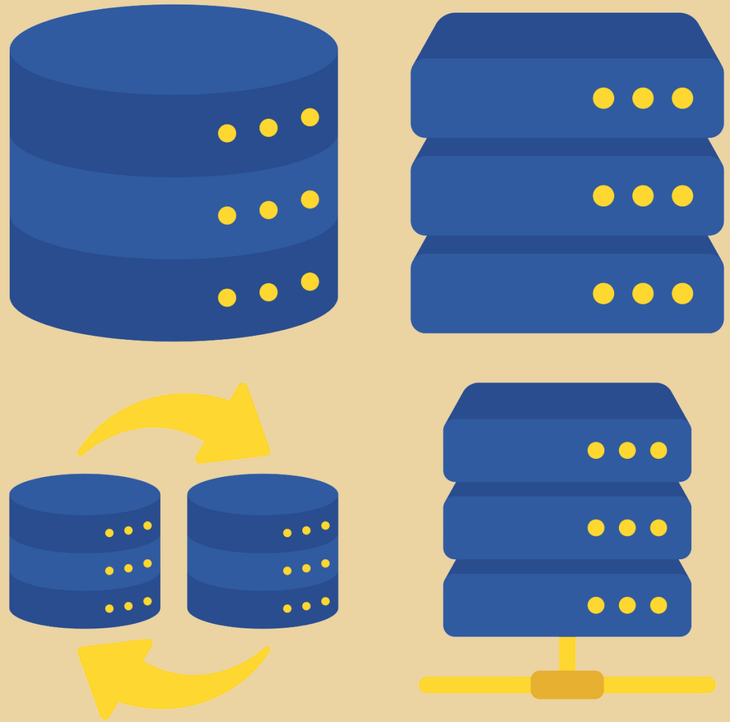
Strong for connected information.

Consistency

Good when every update must be dependable.

Common fits

Payments, orders, inventory, reservations, and school records.



MODEL 02

Flexible by design.

Non-relational databases do not require every record to share one table shape.

Non-relational databases store flexible data

Documents

Self-contained records with fields that can vary.

Key-value pairs

A fast lookup connects one key to one value.

Wide columns

Rows can carry different sets of columns.

Graphs

Nodes and connections make relationships the focus.

The structure can change as the application changes.

WORKED EXAMPLE

Example: a social post can be one flexible document

A post may include text, reactions, tags, images, and optional location data.

Simplified document, not production code.

text — Hiking today

reactions — 24

tags — ["outdoors"]

images — ["trail.jpg"]

location — optional

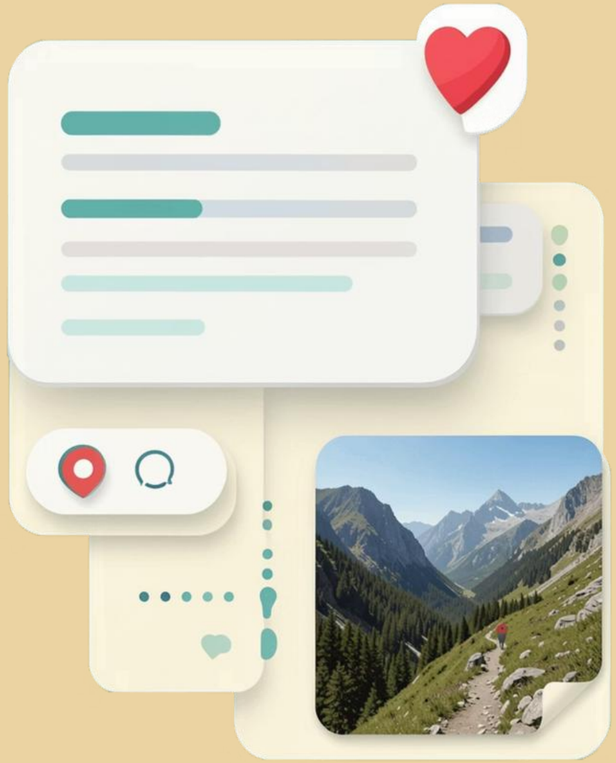
Second example: an online store product catalog

Different products have different attributes, so each product stores only the fields it needs.

laptop — RAM 16 GB, screen 14 in

shoes — size 9, color navy

This flexible structure is why a non-relational database fits the catalog.



WHY NOSQL?

Why choose non-relational databases?

Data varies

Different records need different fields.

Requirements move quickly

The application’s shape is still evolving.

Volume is very high

Many events arrive continuously.

Common fits

Chat, gaming, sensor data, catalogs, sessions, and caching.



SQL VS NOSQL

SQL vs NoSQL: the quick answer

Relational / SQL

- Structured tables with clear relationships
- Strong consistency
- Best for payments, orders, inventory, and school records

Non-relational / NoSQL

- Flexible documents or key-value data
- Data shapes can change
- High scale and speed
- Best for social posts, chat, catalogs, and sensor data

Choose by asking

- Are relationships and accuracy most important? Choose relational.
- Does the data change shape or need flexible scale? Consider non-relational.

Neither is always better—the workload decides, and many systems use both.

Popular SQL and NoSQL database examples

SQL (relational)	NoSQL (non-relational)
PostgreSQL	MongoDB
MySQL	Redis
Oracle Database	Apache Cassandra
Microsoft SQL Server	Amazon DynamoDB
Amazon Aurora	Neo4j

These are popular examples, not a ranking. Each system is designed for different purpose.

SQL vs NoSQL: Key Differences

SQL	NoSQL
Data is stored in tables with rows and columns.	Data is stored as documents, key-value pairs, graphs or columns.
Uses a relational data model.	Uses a non-relational data model.
Has a fixed schema.	Has a flexible schema.
Supports complex joins between tables.	Generally avoids joins.
Follows ACID properties for transactions.	Usually follows the BASE model.
Scales vertically by increasing server capacity.	Scales horizontally by adding more servers.
Best suited for structured data.	Suitable for structured, semi-structured and unstructured data.
Commonly used in banking, ERP and CRM systems.	Commonly used in social media, Big Data and IoT applications.
Examples: MySQL, PostgreSQL	Examples: MongoDB, Cassandra.

Common Questions: SQL or NoSQL?

1. What is the main difference between SQL and NoSQL databases?

2. When should we use SQL, and when should we use NoSQL?

3. Can an application use both SQL and NoSQL databases?

Answers :

1. SQL stores data in structured tables, while NoSQL offers more flexible ways to store data, such as documents and key-value pairs.

2. We should use SQL when data needs clear relationships and high accuracy, such as banking systems. We can use NoSQL when data structures change frequently or we need to handle large amounts of data, such as social media applications.

3. Yes. An application can use both databases. SQL can manage structured data, such as payments, while NoSQL can store flexible data, such as social media posts.

Key Words

SQL

Structured Query Language. A language to ask a database for data.

Relational database

A database that stores data in tables. Tables are linked to each other.

Schema

The plan of a database. It names the tables, the columns, and the type of data in each column.

Document

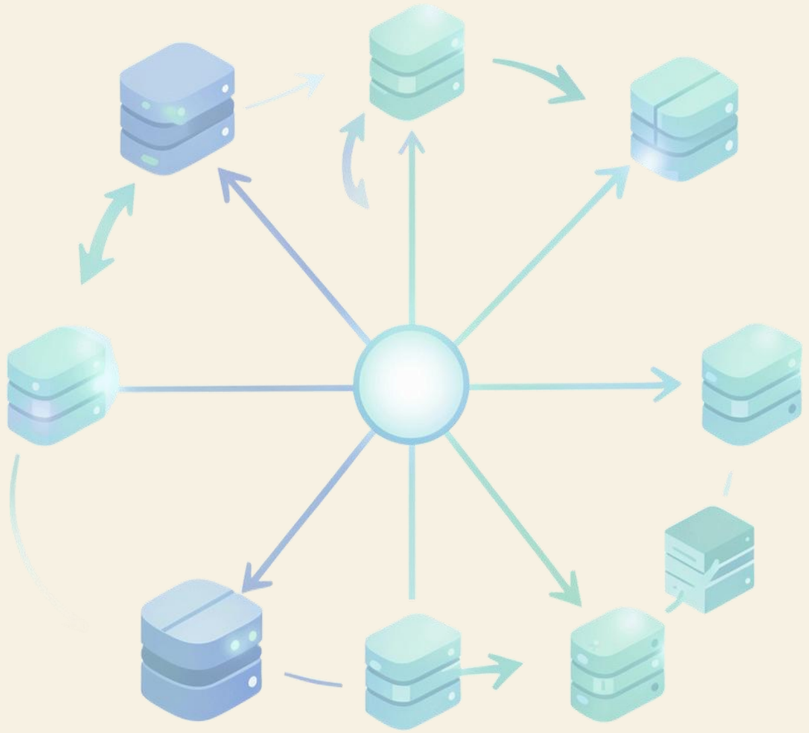
A document is one complete record stored together, like a small digital file. It can hold related details, such as a product name, price, color, and size.

NoSQL

Not only SQL. Databases that do not need fixed tables. They often store flexible documents.

Thanks for listening!

Any Questions?



Sources

- ThoughtSpot
- AWS
- GeekForGeeks